BUS43-3

Valuation & Investment Analysis

DCF, multiples, and the investment thesis

Pricing the firm as judgment under hypothesis sensitivity, not a spreadsheet recipe.

Albert School — 22 hours

Contents

1 DCF valuation

Discounted cash flow values a firm or an investment as the present value of the cash flows it generates, discounted at a rate reflecting their risk. It applies the time-value-of-money and free-cash-flow machinery of BUS33-3 and BUS23-3 to pricing rather than to a single accept/reject decision.

Definition 1 (FCFF and FCFE) *Free Cash Flow to the Firm* is the cash available to all capital providers, before financing flows:

$$\text{FCFF} = \text{EBIT}\ (1-\tau) + \text{D\&A} - \text{capex} - \Delta\ \text{WCR}\ .$$

Free Cash Flow to Equity is the cash available to shareholders after debt flows:

$$\text{FCFE} = \text{FCFF} - \text{interest}\ (1-\tau) + \text{net borrowing}\ .$$

The two are discounted differently and must not be mixed with the wrong rate. FCFF is discounted at the WACC and yields enterprise value; FCFE is discounted at the cost of equity and yields equity value directly.

Definition 2 (DCF value) Over an explicit forecast horizon of n years followed by a terminal value TV_n,

$$\text{EV} = \sum_{t=1}^{n} \frac{\text{FCFF}_t}{(1+\text{WACC})^t} + \frac{\text{TV}_n}{(1+\text{WACC})^n},$$

and analogously for equity value with FCFE discounted at r_e.

Reinvestment discipline. Growth must be paid for. The reinvestment (capex above depreciation plus the increase in working capital) that produces forecast growth must be modelled explicitly and tied to that growth, so that a firm cannot be assumed to grow without consuming the cash that growth requires.

1.1 Terminal value

The terminal value captures all cash flows beyond the explicit horizon. Two standard methods:

Definition 3 (Terminal-value methods)

- *Perpetuity growth (Gordon)*: assuming free cash flow grows at a constant rate $g < \text{WACC}$ forever,

$$\text{TV}_n = \frac{\text{FCFF}_{n+1}}{\text{WACC} - g}.$$

- *Exit multiple*: applying a terminal multiple (e.g. EV/EBITDA) to the final-year metric, $\text{TV}_n = \text{multiple} \times \text{EBITDA}_n$.

Common pitfall Terminal value typically dominates DCF output — often more than half of total enterprise value — because it absorbs the entire post-horizon future. Small changes in g or in the exit multiple move the valuation substantially, so the terminal-value assumption must be stated, justified, and sensitivity-tested, and the perpetuity growth rate must not exceed the long-run growth rate of the economy.

2 Cost of capital revisited

The WACC and CAPM machinery of BUS33-3 is reused, but its inputs are now treated as estimates with honest error bars rather than as point figures.

- **CAPM critiques.** The model rests on assumptions (single-period, mean-variance investors, a single observable market portfolio) that do not hold cleanly; realised returns are a noisy guide to the required return it predicts.
- **Beta estimation problems.** A regression beta depends on the estimation window, the return frequency, and the chosen index; it is unstable over time and may be adjusted toward 1 or sourced from comparable firms (bottom-up beta) and re-levered to the target capital structure.
- **Country-risk adjustments.** For firms exposed to riskier jurisdictions, a country-risk premium is added to the cost of equity (or built into the cash flows), reflecting sovereign and currency risk not captured by a developed- market premium.

Common pitfall The cost of capital is an estimate, not a measurement. Each input carries a range; a WACC quoted to two decimal places claims a precision it does not have. The honest deliverable is a defensible range and a statement of what drives it, not a single number.

3 Trading and market multiples

A multiple prices a company by reference to what the market pays for comparable companies, scaling price by a value driver.

Definition 4 (Standard multiples)

- *EV/EBITDA* and *EV/EBIT* relate *enterprise value* to operating profit, independent of capital structure.
- *P/E* relates *equity price* to net earnings.
- *P/B* relates equity price to book value of equity.

Enterprise-value multiples pair with pre-financing metrics (EBITDA, EBIT); equity-price multiples pair with post-financing metrics (net income, book equity). Mismatching the numerator and denominator (e.g. EV over net income) is an error.

Comparable-set selection. A multiples valuation is only as good as its peer set. Comparables should share the target's business model, growth, margins, and risk; the set must be defended, not assembled by industry label alone.

Common pitfall Multiples mislead when the comparison is not like-for-like:

- **Cyclicality**: a peak-earnings denominator makes a multiple look cheap, a trough-earnings denominator makes it look expensive.
- **Growth differentials**: a faster-growing firm deserves a higher multiple; comparing across growth rates without adjustment misprices.
- **Comparable-set impurity**: peers with different leverage, accounting, or segment mix contaminate the benchmark.

4 Capital markets structure and pricing

Valuation outputs are realised in markets whose structure conditions the price.

Definition 5 (Market structure)

- The *primary market* is where securities are first issued (the firm raises capital); the *secondary market* is where issued securities trade between investors; *OTC* markets trade bilaterally rather than on an exchange.

- **IPO mechanics.** An initial public offering moves a firm from private to public ownership: a price range is set, demand is gathered (book-building), and an offer price is struck — often at a discount to the first trading price.
- **Efficient Market Hypothesis.** The EMH holds that prices reflect available information, so consistent mispricing is hard to exploit. In practice it holds approximately: frictions, limits to arbitrage, and behavioural effects leave room for prices to diverge from intrinsic value.
- **Liquidity discount.** A stake that cannot be sold quickly without moving the price is worth less; private or thinly traded equity carries a discount.
- **Control premium.** A buyer acquiring control pays above the trading price for the ability to direct the firm's cash flows and policy.

5 Non-standard valuations

A default DCF does not fit every business. Each non-standard case requires specific adjustments and metrics, applied at introductory depth.

- **High-growth.** Long, explicit forecast horizons until growth and margins normalise; value concentrates in the terminal year, so terminal assumptions dominate and must be disciplined by a path to steady-state economics.
- **Cyclical.** Earnings must be *normalised* across the cycle rather than taken at a peak or trough; mid-cycle margins drive both DCF and multiples.
- **Distressed.** Going-concern value must be weighed against liquidation value and the probability of default; the cost of capital and capital structure are themselves unstable.

- **Financial institutions.** Debt is raw material, not financing, so FCFF and EV multiples do not apply cleanly; equity cash flows, P/B, and return on equity relative to cost of equity are used instead, with regulatory capital as a constraint.
- **Platform / network-effect.** Value rests on user growth, engagement, and unit economics that precede accounting profit; metrics tracking the network and its monetisation supplement, rather than replace, the cash-flow logic.

6 Scenario, sensitivity, and stress-testing

A valuation is a function of its assumptions, and is reported as such.

> **Definition 6 (Bull / base / bear)** Every valuation deliverable carries three internally consistent cases — a *base* case of most-likely assumptions, a *bull* case and a *bear* case built from coherent upside and downside assumptions — each with testable inputs rather than an arbitrary percentage shift.

Sensitivity analysis isolates the two or three assumptions — typically the growth rate, margins, terminal value, and discount rate — that dominate the output, by varying each and observing the effect on value. Scenario and sensitivity work is a structural discipline running through the valuation, not a table appended at the end.

7 Real-options intuition

A static DCF discounts a single expected path and so ignores the value of managerial flexibility to react as uncertainty resolves.

> **Definition 7 (Real options)** An *option to delay* defers an irreversible commitment until more is known; an *option to abandon* exits and recovers residual value if outcomes disappoint; an *option to expand* scales up if outcomes are favourable.

When a decision is irreversible and the future is uncertain, a DCF point estimate structurally understates value, because it credits neither the downside avoided by waiting or abandoning nor the upside captured by expanding. The intuition is qualitative here: flexibility has value that a single discounted expected path omits.

8 Triangulation across DCF and multiples

DCF and multiples are independent lenses; a valuation reports them together.

> **Definition 8 (Football-field analysis)** A *football-field* chart displays, for one target, the value ranges implied by each method — DCF under its scenarios, each multiple across its peer range — as horizontal bars, making the overlap and the divergences visible.

The output of the valuation is the *reconciliation argument*: the explanation for why the methods diverge — different implied growth, an impure peer set, a terminal-value assumption the market does not share — not the final number or range. A defensible valuation says why the lenses disagree and which to trust here, rather than averaging them.

9 Investment thesis construction

The course terminates in a complete investment thesis on a named public company.

Definition 9 (Investment thesis) A professional *investment memo* states a *buy / hold / sell* recommendation and defends it: the business and its economics, the valuation (DCF and multiples triangulated, with scenarios), the two or three assumptions the conclusion depends on, the risks, and what would change the view.

The recommendation follows from the gap between the triangulated intrinsic value and the market price, sized against the confidence the assumptions can honestly support. The thesis is defended in Q&A against an invited practitioner juror: its strength is measured not by the point estimate but by how well the author knows which inputs drive it and how far it survives their plausible range.